

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY

UNIVERSITY OF INFORMATION TECHNOLOGY

FACULTY OF INFORMATION SYSTEMS



**PROJECT PROPOSAL
TOPIC: AUTHOR COLLABORATOR
FINDER (ARXIV CO-AUTH)**

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CHAPTER 1: INTRODUCTION

1.1. Topic introduction

In the information age, finding potential collaboration partners is a big challenge for scientists, especially in fast-growing fields such as High Energy Physics or Computer Science. This topic focuses on building a co-author recommendation system based on the collaborative network model of researchers on the ArXiv database.

1.2. Urgency and Significance

- Urgency: Help researchers save time, expand their collaborative networks, and increase scientific productivity by connecting with people with relevant expertise.
- Significance: A practical application of Graph Theory and Network Analysis, demonstrating the power of Link Prediction in solving social/collaboration network problems.

CHAPTER 2: DATA ANALYSIS AND NETWORK MODELING

2.1. Dataset

- Dataset Name: ca-* collaboration graphs
- Structure: This is an Undirected Graph:
 - + Nodes/Vertices: Represents Authors.
 - + Edges/Links: Represents Co-authorship (two authors co-wrote a paper).
- Analysis Objective: Analyze network characteristics such as density, degree distribution, and communities to understand the current collaboration structure.

2.2. Project objectives

The specific goal is to build a prototype (demo application) with the function: “When the user enters Author Name X, the system will return a list of K most potential authors to collaborate with, along with justification paths as to why.”

CHAPTER 3: IMPLEMENTATION METHOD

The paper will implement and compare the performance of the following network analysis methods to predict missing links (future collaborative links):

Method	Description (principle)	Role in prediction
1. 2-hop Proximity	Calculate the number of common neighbors or the shortest path distance between two authors X and Y.	Assumption: Two authors with many common co-authors are likely to collaborate.
2. Community Overlap	Identify author communities (research groups) using algorithms like Louvain or Label Propagation, then measure the degree of overlap/same community of X and Y.	Assumption: Authors X and Y are in the same research field(s) (communities) with high potential for collaboration.
3. Link Prediction	Apply graph structure-based metrics (e.g., Adamic-Adar Index, Preferential Attachment) or Machine Learning models on the graph (if scalable).	Give a score to every unaffiliated author pair (X,Y), ranking their potential for collaboration.

CHAPTER 4: PLAN AND DELIVERABLES

4.1. Implementation plan (tentative)

Stage	Job description
Stage 1: Preparation (weeks 1-2)	Load, clean and parse graph data. Analyze basic graph structure (degree, density, connected components).
Stage 2: Model Development (weeks 3-5)	Implement and test 2-hop, Community Overlap, and Link prediction metrics. Evaluate the performance of each method.
Stage 3: Building demo application (weeks 6-7)	Build a simple application interface/functionality (prototype). Integrate the best prediction model. Develop a feature to display justification paths.
Stage 4: Reporting and finalization (week 8)	Write Technical report (10 pages). Prepare slides and complete code.

4.2. Final product

- Code/Demo: Clean, annotated source code (Python, using libraries like NetworkX, scikit-learn, etc.).
- A working demo application (e.g., command line interface or simple web/notebook interface) that implements the suggested functionality.
- Technical report (~10 pages): A detailed technical report in a 1-column format (refer to Springer style), including: introduction, data analysis, methodology description, experimental results, and conclusion.
- Slides: A presentation summarizing the project content.

REFERENCES

- [1] Hassan, D., & Hasan, M. A. (2025). Supervised Link Prediction in Co-Authorship Networks Based on Author Node-Based Features. *arXiv preprint arXiv:2505.21673*.
- [2] Wang, F., Dong, J., Lu, W., & Xu, S. (2023). Collaboration prediction based on multilayer all-author tripartite citation networks: A case study of gene editing. *Journal of Informetrics*, 17(1), 101374.
- [3] Chuan, P. M., Son, L. H., Ali, M., Khang, T. D., Huong, L. T., & Dey, N. (2018). Link prediction in co-authorship networks based on hybrid content similarity metric. *Applied Intelligence*, 48(8), 2470-2486.
- [4] Cho, H., & Yu, Y. (2018). Link prediction for interdisciplinary collaboration via co-authorship network. *Social Network Analysis and Mining*, 8(1), 25.
- [5] Yuliansyah, H., Othman, Z. A., & Bakar, A. A. (2025). Co-authorship prediction method based on degree of gravity and article keywords similarity. *Physica A: Statistical Mechanics and its Applications*, 665, 130511.
- [6] Chennupati, N. (2021). Recommending collaborations using link prediction.
- [7] *Link Property Prediction* (OGB collaboration dataset)